

# Where does bat speed come from, and how much of it becomes exit velocity?

Predicting bat speed for hitters a model has never seen, and checking the lab's bat-ball physics against 2025 MLB data.

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Code: [github.com/Andresperez397/obp-swing-speed](https://github.com/Andresperez397/obp-swing-speed)

## DATA

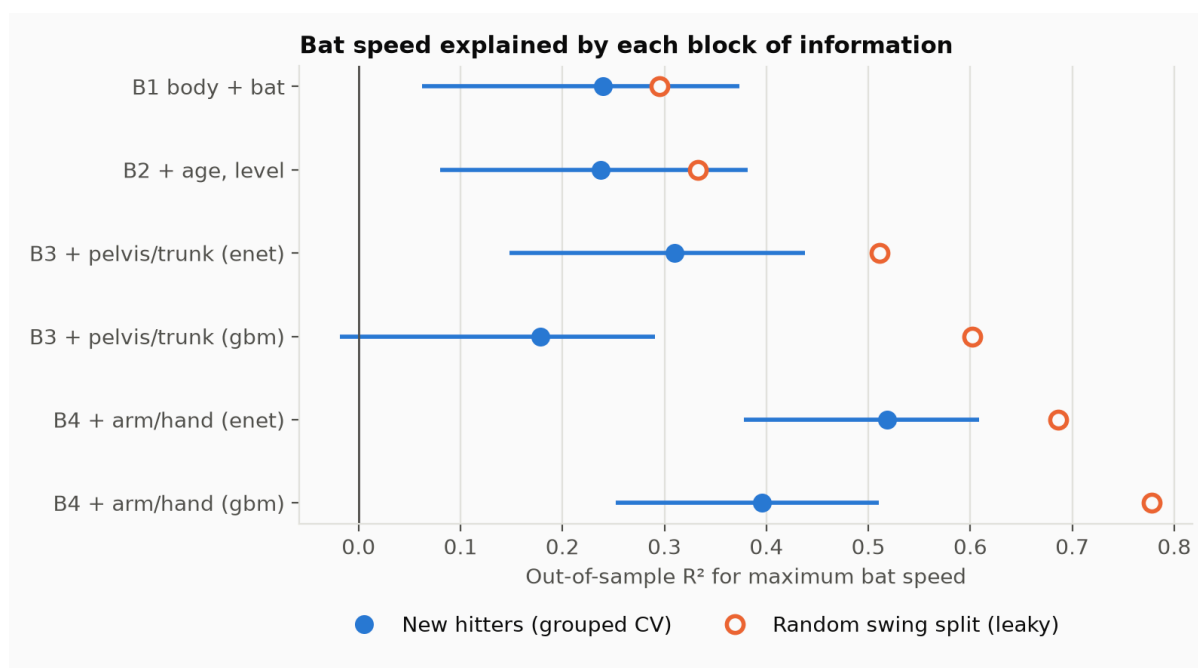
677 swings from 98 hitters in the OpenBiomechanics Project (Driveline Baseball): marker-based motion capture, about 120 swing metrics, exit velocity and HitTrax launch data, hitting off a machine at about 59 mph. The MLB comparison uses 118,793 balls in play with 2025 Statcast bat tracking.

## APPROACH

- **Audit, then a frozen plan:** 22 redundant duplicate variables and 51 invalid pitch speeds found and handled before modeling; plan committed to git first.
- **Unseen hitters:** grouped CV by hitter, 10 folds × 20 repeats; leakage-tested.
- **Fair tuning:** elastic net and boosted trees both tuned by grouped inner CV.
- **Physics:** collision efficiency  $q$  from  $EV = q \cdot v_{\text{pitch}} + (1+q) \cdot v_{\text{bat}}$  (Nathan 2003).

## FINDINGS

- **Bat speed is a hitter trait:** 86% of its variance is between hitters (ICC 0.86).
- **Size gets a quarter of the way:** body and bat size explain  $R^2$  0.24 for new hitters. Pelvis and trunk metrics add +0.06, not separable from zero; arm and hand metrics add +0.20.
- **Simple beat flexible:** tuned boosted trees trailed the elastic net. A leaky random split made them look best (0.78 vs 0.40).
- **Physics checks out:** between hitters, +1 mph bat speed = +1.22 mph exit velocity, matching  $1 + q$ . Within a hitter it's only +0.48.
- **Lab vs MLB:**  $q = 0.236$  in the lab and 0.209 in MLB near squared-up contact. Both are near the textbook value of about 0.2.

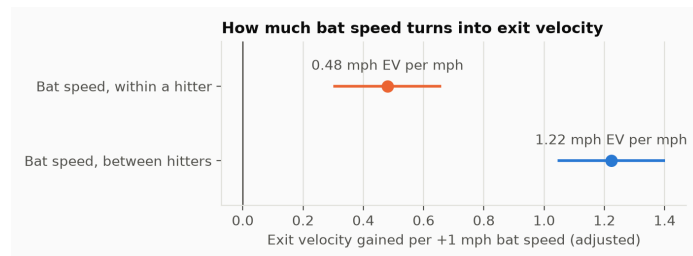


Out-of-sample  $R^2$  for maximum bat speed by information block. Filled: validated on held-out hitters, with 95% CI. Hollow: the same model under a random swing split.

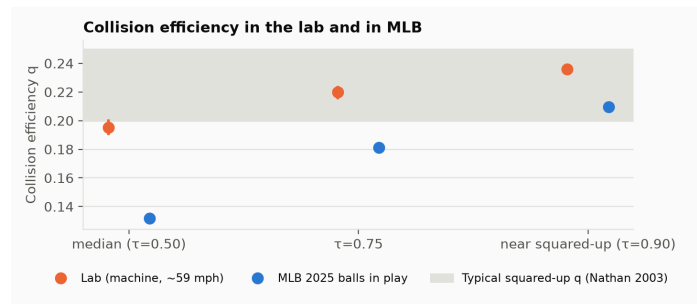
## MODEL RESULTS (HELD-OUT HITTERS)

| Model                                          | R <sup>2</sup> | 95% CI           | RMSE (mph)  | R <sup>2</sup> , leaky split |
|------------------------------------------------|----------------|------------------|-------------|------------------------------|
| Body mass, height, bat size, handedness        | 0.24           | 0.06–0.37        | 4.28        | 0.30                         |
| + age and playing level                        | 0.24           | 0.08–0.38        | 4.28        | 0.33                         |
| + 65 pelvis and trunk metrics (elastic net)    | 0.31           | 0.15–0.44        | 4.08        | 0.51                         |
| <b>+ 27 arm and hand metrics (elastic net)</b> | <b>0.52</b>    | <b>0.38–0.61</b> | <b>3.40</b> | <b>0.69</b>                  |
| Same, tuned boosted trees                      | 0.40           | 0.25–0.51        | 3.81        | 0.78                         |

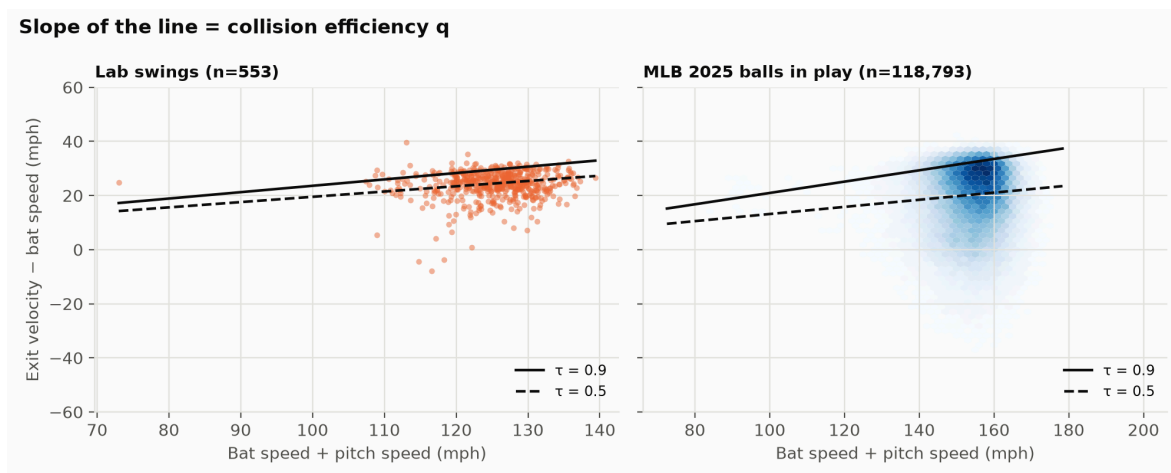
Mean-model RMSE 4.91 mph. R<sup>2</sup> and RMSE: averages over 20 repeats of 10-fold CV grouped by hitter. CIs: hitter bootstrap (2,000 resamples) of the first repeat.



Exit velocity per +1 mph of bat speed, adjusted for attack angle, body mass and handedness.



Collision efficiency  $q$  at three quantiles. MLB intervals are too narrow to see.



## LIMITATIONS

The sample is 98 mostly college hitters at one facility, hitting a machine at about 59 mph, so results may not transfer directly to game swings. The lab bat's material isn't documented, and it affects collision efficiency. The MLB comparison is exploratory: different populations, pitch speeds and measurement systems. With 98 hitters, R<sup>2</sup> gains under about 0.1 aren't separable from zero.

Data: The OpenBiomechanics Project, Driveline Baseball (CC BY-NC-SA 4.0); Baseball Savant / Statcast (MLB Advanced Media). A derived work under the same license, not affiliated with or endorsed by Driveline or MLB.